CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · SHORT NOTES

Chapter 24: Measures of Central Tendency (Mean, Median, Quartiles and Mode)

Topic-wise short notes with solved examples

A measure of central tendency is a single value that represents a whole set of data. This chapter covers the mean, the median, the quartiles and the mode, for raw data and for grouped data.

1 Mean of raw and tabulated data

- **Raw data**: mean $\bar{x} = \dfrac{\sum x}{n}$, the sum of the observations divided by their number.
- **Tabulated data** (values with frequencies): $\bar{x} = \dfrac{\sum fx}{\sum f}$.
- If every observation is increased, decreased, multiplied or divided by the same number, the mean changes in the same way.

Example 1.1 Find the mean of 7, 11, 13, 9, 10.

SOLUTION

$$\bar{x} = \frac{7 + 11 + 13 + 9 + 10}{5} = \frac{50}{5}$$

Mean $= 10$

Example 1.2 The mean of 6 numbers is 15. If one number is removed, the mean of the remaining numbers is 14. Find the number removed.

SOLUTION

Sum of 6 numbers $= 6 \times 15 = 90$; sum of the remaining $5 = 5 \times 14 = 70$

The number removed is $90 - 70 = 20$.

2 Mean of grouped data

- Take the **class mark** x of each class to represent the class.

Method	Formula	Meaning of the symbols
Direct	$\bar{x} = \dfrac{\sum fx}{\sum f}$	x = class mark
Short-cut (assumed mean)	$\bar{x} = A + \dfrac{\sum fd}{\sum f}$	A = assumed mean, $d = x - A$
Step-deviation	$\bar{x} = A + \dfrac{\sum ft}{\sum f} \times i$	$t = \dfrac{x - A}{i}$, i = class width

- All three methods give the same answer. Choose A as a class mark near the middle.

Example 2.1 Find the mean of the following distribution by the step-deviation method.

Class	10–20	20–30	30–40	40–50	50–60
Frequency	4	6	10	12	8

SOLUTION

Take $A = 35$ and $i = 10$.

Class	f	x	$t = \frac{x - 35}{10}$	ft
10–20	4	15	-2	-8
20–30	6	25	-1	-6
30–40	10	35	0	0
40–50	12	45	1	12
50–60	8	55	2	16
Total	40			14

$$\bar{x} = 35 + \frac{14}{40} \times 10 = 35 + 3.5$$

Mean $= 38.5$

3 Median of raw data

- Arrange the data in ascending or descending order. Let the number of observations be n.

n is	Median
Odd	The $\left(\frac{n+1}{2}\right)$th term
Even	The mean of the $\left(\frac{n}{2}\right)$th and $\left(\frac{n}{2}+1\right)$th terms

Example 3.1 Find the median of (i) 12, 3, 20, 7, 18, 9, 15 (ii) 4, 17, 9, 25, 6, 11, 20, 14.

SOLUTION

(i) In order: 3, 7, 9, 12, 15, 18, 20. $n = 7$; the median is the 4th term.

(ii) In order: 4, 6, 9, 11, 14, 17, 20, 25. $n = 8$; the median is the mean of the 4th and 5th terms $= \frac{11 + 14}{2}$.

(i) 12 (ii) 12.5

4 Quartiles of raw data

- The quartiles divide ordered data into four equal parts. Q_1 is the **lower quartile**, Q_2 is the median and Q_3 is the **upper quartile**.

n is	Lower quartile Q_1	Upper quartile Q_3
Odd	The $\left(\frac{n+1}{4}\right)$th term	The $\left(\frac{3(n+1)}{4}\right)$th term
Even	The $\left(\frac{n}{4}\right)$th term	The $\left(\frac{3n}{4}\right)$th term

- **Inter-quartile range** $= Q_3 - Q_1$. **Semi-inter-quartile range** $= \frac{Q_3 - Q_1}{2}$.

Example 4.1 For the data 3, 7, 9, 12, 15, 18, 20, find the quartiles and the inter-quartile range.

SOLUTION

$n = 7$. Q_1 is the $\frac{7+1}{4} = 2$nd term $= 7$.

Q_3 is the $\frac{3 \times 8}{4} = 6$th term $= 18$.

$Q_1 = 7$, $Q_3 = 18$ **and inter-quartile range** $= 11$

Example 4.2 For the data 4, 6, 9, 11, 14, 17, 20, 25, find the quartiles and the semi-inter-quartile range.

SOLUTION

$n = 8$. Q_1 is the $\frac{8}{4} = 2$nd term $= 6$.

Q_3 is the $\frac{3 \times 8}{4} = 6$th term $= 17$.

$Q_1 = 6$, $Q_3 = 17$ **and semi-inter-quartile range** $= \frac{17 - 6}{2} = 5.5$

5 Median and quartiles of grouped data

- Draw the **ogive** (cumulative frequency curve) for the distribution, with total frequency N.
- Read the median at cumulative frequency $\frac{N}{2}$, Q_1 at $\frac{N}{4}$ and Q_3 at $\frac{3N}{4}$.
- For values with frequencies (not classes), use the cumulative frequency column to find the term in the required position.

Example 5.1 Find the median of this distribution.

Value	5	6	7	8	9
Frequency	3	8	12	9	5

SOLUTION

Cumulative frequencies: 3, 11, 23, 32, 37. $n = 37$, which is odd.

The median is the $\frac{37+1}{2} = 19$th term. The 12th to the 23rd terms are all 7.

Median $= 7$

6 Mode

- The **mode** is the value that occurs most often. For tabulated data it is the value with the highest frequency.
- A set of data can have more than one mode, or no mode.
- For grouped data, the class with the highest frequency is the **modal class**, and the mode is found from the **histogram**.

Example 6.1 Find the mode of 4, 7, 5, 7, 9, 7, 5, 4, 7, and of the distribution in the previous example.

SOLUTION

In the list, 7 occurs four times, more than any other value.

In the distribution, the highest frequency is 12, for the value 7.

The mode is 7 in both.

7 Common mistakes

- Finding the median without first arranging the data in order.
- Using the class limits instead of the class marks for the mean.
- Forgetting to multiply by the class width i in the step-deviation method.
- Giving the position of the median instead of the term in that position.